

SS-RESfM Results Through VGPA's Five Lenses — Complete

Lenses: per-scene wins / mean vs. median + blow-ups / coverage / RA-AUC + AUC / rotation
plus our sixth lens (seed reliability) and the contamination axis — September 12, 2026

Reading guide

All five VGPA lenses applied to our full 5-seed matrix, without VGPA rows (their numbers are cross-track/cross-environment; see the separate comparison note). Lens 4 is complete: AUC@30 (registered pairs only) vs. RA-AUC@30 (all pairs, unregistered = 180°) — the pair quantifies exactly how much each method's raw accuracy is bought with coverage. The seed-reliability lens and the contamination-axis section cover the axes VGPA does not measure. In-distribution RA-AUC bands are computing (fleet in flight).

megadepth (Nc=13126)

Method	Trans mean±std	Trans med (pooled)	Rot mean	Nr%	seeds
SS madweight	0.512±0.08	0.151	3.70	81%	5
SS weight	0.512±0.09	0.188	3.28	79%	5
SS weight+TTT	0.398±0.11	0.175	2.88	77%	5
SS remove	0.512±0.08	0.117	4.04	72%	5
SS remove+TTT	0.561±0.07	0.143	4.22	71%	5
SS hybrid	0.455±0.09	0.179	3.06	77%	5
RESfM-scratch (sup.)	0.365±0.12	0.068	1.89	85%	5
ESFM (no mech.)	0.714±0.11	0.297	6.03	77%	5
ESFM* (clean tracks)	0.594±0.18	0.071	6.18	90%	5

Per-scene wins vs RESfM-scratch (matched seed, scene x seed cells) SS madweight: trans 75/180 (42%), rot 63/180 (35%)

SS weight: trans 55/180 (31%), rot 44/180 (24%)

SS weight+TTT: trans 66/180 (37%), rot 53/180 (29%)

SS remove: trans 73/180 (41%), rot 62/180 (34%)

SS remove+TTT: trans 67/180 (37%), rot 65/180 (36%)

SS hybrid: trans 66/179 (37%), rot 57/179 (32%)

ESFM (no mech.): trans 47/180 (26%), rot 44/180 (24%)

ESFM* (clean tracks): trans 105/180 (58%), rot 107/180 (59%)

Mean-driving scenes (seed-20, top 3 by scene mean) SS madweight (med 0.14): 0265 3.26, 0147 2.99, 0130 1.60

SS weight (med 0.18): 0099 3.52, 0265 2.34, 0083 1.56

SS weight+TTT (med 0.19): 0455 1.60, 0147 1.20, 0024 1.13

SS remove (med 0.19): 0265 3.04, 0076 3.04, 0147 2.96

SS remove+TTT (med 0.13): 5016 3.10, 0147 3.02, 0060 2.52

SS hybrid (med 0.20): 0265 2.43, 0083 1.79, 0197 1.34

RESfM-scratch (sup.) (med 0.07): 0147 3.22, 0265 3.09, 0024 1.09

ESFM (no mech.) (med 0.31): 0265 3.01, 0046 2.90, 0102 2.28

ESFM* (clean tracks) (med 0.06): 0147 4.18, 0102 3.40, 0265 3.16

1dsfm (Nc=4563)

Method	Trans mean $\pm$ std	Trans med (pooled)	Rot mean	Nr%	seeds
SS madweight	11.041 $\pm$ 1.68	3.956	7.74	70%	5
SS weight	14.349 $\pm$ 2.97	11.102	8.24	79%	5
SS weight+TTT	15.719 $\pm$ 2.10	11.205	10.09	81%	5
SS remove	9.125 $\pm$ 2.17	1.615	6.94	58%	5
SS remove+TTT	8.925 $\pm$ 1.20	1.749	6.90	58%	5
SS hybrid	16.232 $\pm$ 2.20	10.181	7.89	80%	5
RESfM-scratch (sup.)	10.498 $\pm$ 1.45	2.558	9.62	77%	5
ESFM (no mech.)	18.415 $\pm$ 0.75	13.409	14.53	84%	5
ESFM* (clean tracks)	13.236 $\pm$ 0.52	2.951	13.29	80%	5

Per-scene wins vs RESfM-scratch (matched seed, scene x seed cells) SS madweight: trans 22/50 (44%), rot 31/50 (62%)

SS weight: trans 18/50 (36%), rot 29/50 (58%)

SS weight+TTT: trans 16/50 (32%), rot 26/50 (52%)

SS remove: trans 33/50 (66%), rot 39/50 (78%)

SS remove+TTT: trans 35/50 (70%), rot 39/50 (78%)

SS hybrid: trans 15/50 (30%), rot 27/50 (54%)

ESFM (no mech.): trans 10/50 (20%), rot 17/50 (34%)

ESFM* (clean tracks): trans 24/50 (48%), rot 26/50 (52%)

Mean-driving scenes (seed-20, top 3 by scene mean) SS madweight (med 4.37): Tower_of_London 76.34, Ellis_Island 17.59, Madrid_Metropolis 9.63

SS weight (med 11.01): Yorkminster 30.87, Tower_of_London 28.29, Vienna_Cathedral 22.71

SS weight+TTT (med 7.16): Tower_of_London 72.61, Vienna_Cathedral 33.57, Ellis_Island 17.11

SS remove (med 1.78): Tower_of_London 36.99, Yorkminster 16.98, Ellis_Island 16.95

SS remove+TTT (med 3.42): Tower_of_London 36.59, Ellis_Island 15.72, Yorkminster 10.25

SS hybrid (med 4.25): Tower_of_London 71.81, Yorkminster 30.45, Ellis_Island 17.12

RESfM-scratch (sup.) (med 2.93): Tower_of_London 67.95, Ellis_Island 16.02, Yorkminster 8.77

ESFM (no mech.) (med 14.37): Tower_of_London 59.13, Madrid_Metropolis 35.77, Ellis_Island 25.12

ESFM* (clean tracks) (med 4.29): Tower_of_London 69.64, Yorkminster 22.41, Vienna_Cathedral 17.23

1dsfmhard (Nc=1497)

Method	Trans mean $\pm$ std	Trans med (pooled)	Rot mean	Nr%	seeds
SS madweight	18.918 $\pm$ 4.05	13.552	27.65	43%	5
SS weight	22.924 $\pm$ 2.79	13.134	27.95	40%	5
SS weight+TTT	22.242 $\pm$ 3.17	13.305	30.60	41%	5
SS remove	12.419 $\pm$ 2.05	11.853	12.27	29%	5
SS remove+TTT	15.150 $\pm$ 0.98	11.299	16.40	32%	5
SS hybrid	22.983 $\pm$ 4.02	13.150	28.21	41%	5
RESfM-scratch (sup.)	19.158 $\pm$ 3.42	13.714	27.30	34%	5
ESFM (no mech.)	21.758 $\pm$ 0.80	24.280	32.01	40%	5
ESFM* (clean tracks)	18.121 $\pm$ 4.35	15.248	21.55	36%	5

Per-scene wins vs RESfM-scratch (matched seed, scene x seed cells) SS madweight: trans 12/25 (48%), rot 17/25 (68%)

SS weight: trans 12/25 (48%), rot 14/25 (56%)

SS weight+TTT: trans 14/25 (56%), rot 15/25 (60%)

SS remove: trans 20/25 (80%), rot 21/25 (84%)

SS remove+TTT: trans 18/25 (72%), rot 20/25 (80%)

SS hybrid: trans 12/25 (48%), rot 17/25 (68%)

ESFM (no mech.): trans 11/25 (44%), rot 9/25 (36%)

ESFM* (clean tracks): trans 13/25 (52%), rot 13/25 (52%)

Mean-driving scenes (seed-20, top 3 by scene mean) SS madweight (med 8.82): Gendarmenmarkt 31.17, Union_Square 13.55, Roman_Forum 8.82

SS weight (med 13.13): Roman_Forum 40.77, Gendarmenmarkt 33.15, Union_Square 13.13

SS weight+TTT (med 17.67): Roman_Forum 64.97, Gendarmenmarkt 29.30, Trafalgar 17.67

SS remove (med 11.31): Roman_Forum 21.09, Gendarmenmarkt 15.51, Union_Square 11.31

SS remove+TTT (med 11.30): Gendarmenmarkt 31.05, Roman_Forum 28.77, Union_Square 11.30

SS hybrid (med 13.15): Roman_Forum 64.16, Gendarmenmarkt 32.27, Union_Square 13.15

RESfM-scratch (sup.) (med 13.08): Gendarmenmarkt 33.20, Union_Square 13.35, Roman_Forum 13.08

ESFM (no mech.) (med 24.57): Gendarmenmarkt 31.79, Roman_Forum 29.57, Trafalgar 24.57

ESFM* (clean tracks) (med 16.23): Gendarmenmarkt 22.71, Trafalgar 17.07, Roman_Forum 16.23

strecha (Nc=54)

Method	Trans mean $\pm$ std	Trans med (pooled)	Rot mean	Nr%	seeds
SS madweight	2.650 $\pm$ 0.25	0.938	12.80	91%	5
SS weight	1.969 $\pm$ 0.09	0.006	7.61	98%	5
SS weight+TTT	2.000 $\pm$ 0.05	0.006	7.95	99%	5
SS remove	3.201 $\pm$ 0.11	2.034	14.39	97%	5
SS remove+TTT	3.008 $\pm$ 0.07	1.796	13.23	96%	5
SS hybrid	3.254 $\pm$ 0.45	2.704	15.15	71%	5
RESfM-scratch (sup.)	0.388 $\pm$ 0.72	0.005	1.56	99%	5
ESFM (no mech.)	2.034 $\pm$ 0.14	0.005	7.61	100%	5
ESFM* (clean tracks)	1.139 $\pm$ 0.45	0.007	11.13	99%	5

Per-scene wins vs RESfM-scratch (matched seed, scene x seed cells) SS madweight: trans 0/20 (0%), rot 1/20 (5%)

SS weight: trans 3/20 (15%), rot 8/20 (40%)

SS weight+TTT: trans 3/20 (15%), rot 9/20 (45%)

SS remove: trans 0/20 (0%), rot 0/20 (0%)

SS remove+TTT: trans 0/20 (0%), rot 0/20 (0%)

SS hybrid: trans 0/20 (0%), rot 0/20 (0%)

ESFM (no mech.): trans 3/20 (15%), rot 8/20 (40%)

ESFM* (clean tracks): trans 3/20 (15%), rot 7/20 (35%)

Mean-driving scenes (seed-20, top 3 by scene mean) SS madweight (med 0.97): entry-P10 8.04, fountain-P11 1.57, Herz-Jesu-P25 0.37

SS weight (med 0.03): entry-P10 7.96, Herz-Jesu-P25 0.05, Herz-Jesu-P8 0.01

SS weight+TTT (med 0.01): entry-P10 7.89, Herz-Jesu-P25 0.01, Herz-Jesu-P8 0.01

SS remove (med 1.68): entry-P10 8.72, Herz-Jesu-P25 1.79, fountain-P11 1.56

SS remove+TTT (med 1.56): entry-P10 8.75, fountain-P11 1.92, Herz-Jesu-P25 1.20

SS hybrid (med 1.39): entry-P10 8.02, fountain-P11 2.75, Herz-Jesu-P25 0.03

RESfM-scratch (sup.) (med 0.01): entry-P10 0.01, Herz-Jesu-P25 0.01, fountain-P11 0.01

ESFM (no mech.) (med 0.01): entry-P10 7.84, Herz-Jesu-P25 0.01, Herz-Jesu-P8 0.01

ESFM* (clean tracks) (med 0.95): fountain-P11 2.72, entry-P10 1.90, Herz-Jesu-P25 0.01

blendedmvs (Nc=225)

Method	Trans mean $\pm$ std	Trans med (pooled)	Rot mean	Nr%	seeds
SS madweight	0.140 $\pm$ 0.02	0.004	9.01	92%	5
SS weight	0.144 $\pm$ 0.03	0.015	9.17	94%	5
SS weight+TTT	0.137 $\pm$ 0.02	0.002	8.30	94%	5
SS remove	0.240 $\pm$ 0.06	0.160	16.04	88%	5
SS remove+TTT	0.219 $\pm$ 0.10	0.211	13.19	90%	5
SS hybrid	0.140 $\pm$ 0.02	0.004	9.01	93%	5
RESfM-scratch (sup.)	0.353 $\pm$ 0.04	0.260	31.66	92%	5
ESFM (no mech.)	6.449 $\pm$ 3.75	0.000	12.80	96%	5
ESFM* (clean tracks)	3.795 $\pm$ 3.51	0.000	10.62	94%	5

Per-scene wins vs RESfM-scratch (matched seed, scene x seed cells) SS madweight: trans 10/20 (50%), rot 11/20 (55%)

SS weight: trans 11/20 (55%), rot 11/20 (55%)

SS weight+TTT: trans 10/20 (50%), rot 10/20 (50%)

SS remove: trans 6/20 (30%), rot 10/20 (50%)

SS remove+TTT: trans 11/20 (55%), rot 12/20 (60%)

SS hybrid: trans 11/20 (55%), rot 10/20 (50%)

ESFM (no mech.): trans 3/8 (38%), rot 3/8 (38%)

ESFM* (clean tracks): trans 3/8 (38%), rot 2/8 (25%)

Mean-driving scenes (seed-20, top 3 by scene mean) SS madweight (med 0.00): 5b950c71608de421b1e7318f 0.58, 5acf8ca0f3d8a750097e4b15 0.01, 5a48ba95c7dab83a7d7b44ed 0.00

SS weight (med 0.04): 5b950c71608de421b1e7318f 0.33, 5a48ba95c7dab83a7d7b44ed 0.05, 5acf8ca0f3d8a750097e4b15 0.02

SS weight+TTT (med 0.01): 5b950c71608de421b1e7318f 0.52, 5acf8ca0f3d8a750097e4b15 0.02, 5a48ba95c7dab83a7d7b44ed 0.00

SS remove (med 0.19): 5a48ba95c7dab83a7d7b44ed 0.72, 5b950c71608de421b1e7318f 0.33, 5acf8ca0f3d8a750097e4b15 0.05

SS remove+TTT (med 0.07): 5a48ba95c7dab83a7d7b44ed 0.62, 5b950c71608de421b1e7318f 0.13, 5acf8ca0f3d8a750097e4b15 0.00

SS hybrid (med 0.03): 5b950c71608de421b1e7318f 0.37, 5acf8ca0f3d8a750097e4b15 0.03, 5a48ba95c7dab83a7d7b44ed 0.02

RESfM-scratch (sup.) (med 0.35): 5b950c71608de421b1e7318f 0.82, 5a48ba95c7dab83a7d7b44ed 0.69, 5acf8ca0f3d8a750097e4b15 0.00

ESFM (no mech.) (med 0.22): 5b950c71608de421b1e7318f 0.47, 5a48ba95c7dab83a7d7b44ed 0.44, 5acf8ca0f3d8a750097e4b15 0.00

ESFM* (clean tracks) (med 0.27): 5b950c71608de421b1e7318f 0.76, 5a48ba95c7dab83a7d7b44ed 0.54, 5acf8ca0f3d8a750097e4b15 0.00

olsson (Nc=3858)

Method	Trans mean $\pm$ std	Trans med (pooled)	Rot mean	Nr%	seeds
SS madweight	3.010 $\pm$ 0.34	1.169	14.44	86%	5
SS weight	2.906 $\pm$ 0.33	1.139	13.24	90%	5
SS weight+TTT	8.033 $\pm$ 0.30	0.433	11.60	90%	5
SS remove	3.413 $\pm$ 0.57	1.619	13.89	78%	5
SS remove+TTT	3.626 $\pm$ 0.35	1.913	14.53	78%	5
SS hybrid	2.905 $\pm$ 0.23	1.559	13.35	88%	5
RESfM-scratch (sup.)	7.374 $\pm$ 2.63	1.123	12.62	89%	5
ESFM (no mech.)	5.914 $\pm$ 3.32	0.558	13.03	90%	5
ESFM* (clean tracks)	9.363 $\pm$ 1.28	1.459	18.95	90%	5

Per-scene wins vs RESfM-scratch (matched seed, scene x seed cells) SS madweight: trans 81/195 (42%), rot 92/195 (47%)
SS weight: trans 87/195 (45%), rot 98/195 (50%)
SS weight+TTT: trans 110/195 (56%), rot 102/195 (52%)
SS remove: trans 76/195 (39%), rot 82/195 (42%)
SS remove+TTT: trans 71/195 (36%), rot 81/195 (42%)
SS hybrid: trans 85/194 (44%), rot 92/194 (47%)
ESFM (no mech.): trans 107/195 (55%), rot 99/195 (51%)
ESFM* (clean tracks): trans 88/195 (45%), rot 77/195 (39%)

Mean-driving scenes (seed-20, top 3 by scene mean) SS madweight (med 1.15): Lund_University_Sphinx 14.42, Jonas_Ahlstromer 10.89, Tsar_Nikolai_I 10.50
SS weight (med 1.22): Urban_II 12.88, GustavIIAdolf 10.10, Lund_University_Sphinx 9.96
SS weight+TTT (med 0.18): dtu106 236.90, Urban_II 12.25, Porta_San_Donato_Bologna 7.28
SS remove (med 1.39): Tsar_Nikolai_I 10.55, Urban_II 9.26, Lund_University_Sphinx 8.84
SS remove+TTT (med 2.26): Statue_Of_Liberty 20.74, Tsar_Nikolai_I 15.24, Urban_II 13.68
SS hybrid (med 1.34): Urban_II 14.01, Lund_University_Sphinx 9.43, Park_Gate_Clermont_Ferrand 9.06
RESfM-scratch (sup.) (med 0.17): dtu106 239.16, Lund_University_Sphinx 11.51, Tsar_Nikolai_I 10.75
ESFM (no mech.) (med 0.13): dtu106 236.32, Statue_Of_Liberty 11.98, Urban_II 11.66
ESFM* (clean tracks) (med 0.28): dtu106 238.99, Tsar_Nikolai_I 16.31, Statue_Of_Liberty 15.69
(RA-AUC/AUC: pending the Final_recon pose-recovery program.)

Lens 4 complete: RA-AUC@30 / AUC@30 (pose), 5-seed bands, OOD matrix

Method	megadepth	1dsfm	1dsfmhard	strecha	blendedmvs	olsson
<i>RA-AUC@30 (coverage-priced)</i>						
SS madweight	0.599±0.02	0.374±0.01	0.067±0.01	0.587±0.03	0.589±0.07	0.568±0.01
SS weight	0.582±0.02	0.444±0.02	0.092±0.01	0.633±0.04	0.594±0.02	0.626±0.02
SS weight+TTT	0.580±0.02	0.445±0.02	0.092±0.01	0.635±0.03	0.595±0.02	0.627±0.03
SS remove	0.442±0.01	0.259±0.01	0.036±0.01	0.612±0.02	0.500±0.08	0.468±0.02
SS remove+TTT	0.442±0.01	0.262±0.01	0.037±0.01	0.609±0.02	0.501±0.08	0.467±0.02
SS hybrid	0.559±0.03	0.442±0.02	0.095±0.01	0.438±0.07	0.594±0.02	0.531±0.02
RESfM-scratch (sup.)	0.605±0.01	0.408±0.01	0.044±0.01	0.725±0.08	0.600±0.01	0.620±0.01
ESFM (no mech.)	0.574±0.02	0.461±0.01	0.099±0.01	0.662±0.03	0.722±0.06	0.645±0.01
ESFM* (clean)	0.691±0.04	0.461±0.02	0.035±0.01	0.517±0.03	0.652±0.10	0.570±0.14
<i>AUC@30 (registered pairs only)</i>						
SS madweight	0.778±0.02	0.642±0.03	0.354±0.06	0.601±0.03	0.642±0.08	0.640±0.02
SS weight	0.777±0.01	0.582±0.02	0.408±0.04	0.648±0.04	0.635±0.03	0.670±0.02
SS weight+TTT	0.777±0.01	0.578±0.02	0.411±0.03	0.650±0.04	0.637±0.03	0.674±0.02
SS remove	0.755±0.00	0.623±0.02	0.411±0.07	0.636±0.06	0.577±0.05	0.584±0.01
SS remove+TTT	0.748±0.00	0.638±0.02	0.388±0.03	0.634±0.06	0.578±0.05	0.581±0.02
SS hybrid	0.760±0.01	0.573±0.02	0.399±0.04	0.603±0.05	0.637±0.03	0.644±0.01
RESfM-scratch (sup.)	0.811±0.01	0.623±0.03	0.329±0.06	0.735±0.07	0.654±0.02	0.653±0.02
ESFM (no mech.)	0.756±0.01	0.570±0.01	0.409±0.04	0.677±0.05	0.744±0.05	0.688±0.01
ESFM* (clean)	0.790±0.05	0.669±0.04	0.330±0.07	0.519±0.03	0.760±0.06	0.607±0.15

Seed-reliability lens (unique to us): seed-mean range + catastrophic cells

Catastrophic cell = scene x seed with translation mean > 30 (or >100 marked separately).

Method	megadepth range / cat	1dsfm range / cat	1dsfmhard range / cat	strecha range / cat	blendedmvs range / cat	olsson range / cat
SS madweight	[0.42,0.59] 0/180	[8.84,12.78] 4/50	[12.21,22.17] 7/25	[2.33,2.98] 0/20	[0.11,0.16] 0/20	[2.57,3.37] 0/195
SS weight	[0.42,0.63] 0/180	[12.15,19.03] 6/50	[18.49,25.21] 8/25	[1.88,2.12] 0/20	[0.10,0.18] 0/20	[2.52,3.33] 0/195
SS weight+TTT	[0.32,0.57] 0/180	[12.71,18.45] 7/50	[19.24,26.44] 5/25	[1.96,2.09] 0/20	[0.11,0.16] 0/20	[7.74,8.48] 5/195 (5>100)
SS remove	[0.38,0.60] 0/180	[6.34,12.00] 5/50	[10.64,15.76] 2/25	[3.02,3.33] 0/20	[0.17,0.30] 0/20	[2.81,4.35] 0/195
SS remove+TTT	[0.49,0.66] 0/180	[7.99,10.50] 5/50	[13.61,16.34] 5/25	[2.94,3.11] 0/20	[0.07,0.36] 0/20	[3.23,4.07] 0/195
SS hybrid	[0.37,0.56] 0/179	[13.84,19.35] 9/50	[17.31,26.35] 8/25	[2.70,3.83] 0/20	[0.10,0.16] 0/20	[2.64,3.16] 0/194
RESfM-scratch (sup.)	[0.19,0.53] 0/180	[8.52,12.47] 5/50	[16.22,24.78] 6/25	[0.01,1.67] 0/20	[0.28,0.39] 0/20	[2.69,8.86] 4/195 (4>100)
ESFM (no mech.)	[0.58,0.87] 0/180	[17.44,19.41] 10/50	[20.37,22.30] 6/25	[1.84,2.18] 0/20	[0.23,9.77] 3/20	[1.95,8.52] 3/195 (3>100)
ESFM* (clean)	[0.45,0.91] 0/180	[12.57,13.84] 5/50	[13.68,23.98] 2/25	[0.62,1.84] 0/20	[0.00,8.08] 1/20	[8.56,11.64] 5/195 (5>100)

Contamination axis (in-distribution matrix), lenses 1-3+5

olssonid (train@0.5%)

Recipe	Trans mean $\pm$ std	Trans med	Rot	Nr (abs)	seeds
SUP	2.03 $\pm$ 1.86	0.06	12.1	849	5
madweight	1.97 $\pm$ 0.94	0.61	6.6	721	5
weight	3.05 $\pm$ 1.39	1.96	11.5	812	5
weight+TTT	1.60 $\pm$ 0.81	0.26	6.6	820	5
remove	2.94 $\pm$ 0.27	3.12	11.2	624	5
vanilla	1.45 $\pm$ 1.44	0.02	7.2	829	5

Per-scene wins vs SUP (matched seed): madweight: trans 17/40 (42%), rot 16/40 (40%)

weight: trans 14/40 (35%), rot 14/40 (35%)

weight+TTT: trans 21/40 (52%), rot 19/40 (48%)

remove: trans 14/40 (35%), rot 14/40 (35%)

vanilla: trans 29/40 (72%), rot 30/40 (75%)

bmvsid (train@3.1%)

Recipe	Trans mean $\pm$ std	Trans med	Rot	Nr (abs)	seeds
SUP	0.26 $\pm$ 0.15	0.00	20.7	207	5
madweight	0.09 $\pm$ 0.10	0.01	4.2	207	5
weight	0.11 $\pm$ 0.01	0.00	7.3	208	5
weight+TTT	0.08 $\pm$ 0.05	0.00	5.6	200	5
remove	0.23 $\pm$ 0.08	0.24	13.2	190	5
vanilla	2.36 $\pm$ 3.27	0.00	3.5	205	5

Per-scene wins vs SUP (matched seed): madweight: trans 7/20 (35%), rot 7/20 (35%)

weight: trans 7/20 (35%), rot 7/20 (35%)

weight+TTT: trans 7/20 (35%), rot 7/20 (35%)

remove: trans 8/20 (40%), rot 8/20 (40%)

vanilla: trans 0/5 (0%), rot 0/5 (0%)

1donly (train@43%)

Recipe	Trans mean $\pm$ std	Trans med	Rot	Nr (abs)	seeds
SUP	4.97 $\pm$ 2.08	1.29	6.0	856	5
madweight	6.37 $\pm$ 0.20	1.72	7.1	883	5
weight	10.65 $\pm$ 2.49	12.81	11.8	942	5
weight+TTT	9.91 $\pm$ 2.39	10.49	11.0	932	5
remove	6.68 $\pm$ 1.44	1.17	6.1	767	5
vanilla	10.20 $\pm$ 2.97	9.90	12.9	1014	5

Per-scene wins vs SUP (matched seed): madweight: trans 6/15 (40%), rot 7/15 (47%)

weight: trans 1/15 (7%), rot 4/15 (27%)

weight+TTT: trans 1/15 (7%), rot 4/15 (27%)

remove: trans 6/15 (40%), rot 10/15 (67%)

vanilla: trans 1/15 (7%), rot 4/15 (27%)

hardonly (train@60%)

Recipe	Trans mean $\pm$ std	Trans med	Rot	Nr (abs)	seeds
SUP	23.38 $\pm$ 7.76	16.80	12.4	106	5
madweight	12.13 $\pm$ 6.12	10.56	10.2	226	5
weight	28.11 $\pm$ 5.94	22.87	31.5	316	5
weight+TTT	29.27 $\pm$ 8.44	26.87	32.8	321	5
remove	19.42 $\pm$ 7.72	13.54	10.9	120	5
vanilla	23.14 $\pm$ 5.89	20.64	33.3	301	5

Per-scene wins vs SUP (matched seed): madweight: trans 6/10 (60%), rot 7/10 (70%)

weight: trans 3/10 (30%), rot 2/10 (20%)

weight+TTT: trans 1/10 (10%), rot 1/10 (10%)

remove: trans 6/10 (60%), rot 7/10 (70%)

vanilla: trans 4/10 (40%), rot 1/10 (10%)

(Lens 4 for the in-dist matrix: fleet running; RA-AUC bands land next check.)